1.)

If $x^2 + 5x$ is odd then x is odd.

Contrapositive: if x is even then $x^2 + 5x$ is even.

A number n is even iff $2k = n$.

Since we're assuming x is even, we can write it as $2k$.

Likewise with $x^2 + 5x \Rightarrow (2k)^2 + 5(2k) \rightarrow 4k^2 + 10k$

$2\underbrace{(2k^2 + 5k)}_{\text{int.}} \leftarrow$ Definitionally even.

We assumed x is even $(2k)$, and $x^2 + 5x$ also even $(2k)^2 + 5(2k)$ which proves true.

□ Since the contrapositive is true, the original claim is true.

2.)

Given $x_1 < x_2$, if $\underbrace{f(x_1) > f(x_2)}_{p}$ then $\underbrace{f(x) \text{ is not increasing}}_{q}$.

$\sim q \Rightarrow \sim p =$ Given $x_1 < x_2$, if $f(x)$ is increasing then $f(x_1) < f(x_2)$.

★ $f(x)$ is increasing on domain $\mathbb{R}$ iff $\forall x \in \mathbb{R},\ \forall k > 0,\ f(x) < f(x + k)$.

Let $x_2 - x_1 = k$. k here is > 0 since $x_1 < x_2$.

Since $k = x_2 - x_1$, we can rewrite it as

$x_2 = x_1 + k$

Using the definition, substituting $x = x_1$ and $k = x_2 - x_1$, so $x + k = x_2$

$f(x_1) < f(x_2)$

□ Therefore, given $x_1 < x_2$, when $f(x)$ is increasing then $f(x_1) < f(x_2)$.

3.)

If $\underbrace{f(x) = x^2}_{p}$ then $\underbrace{f(x) \text{ is not bounded above}}_{q}$.

$p \rightarrow \sim q =$ if $f(x) = x^2$ then $f(x)$ is bounded above.

★ $f(x)$ is bounded above iff $\exists K \in \mathbb{R},\ \forall x \in \mathbb{R},\ f(x) \leq K$

Since $f(x)$ is x^2 in this case, we can sub x^2 for $f(x)$ in our definition:

$\exists K \in \mathbb{R},\ \forall x \in \mathbb{R},\ x^2 \leq K$.

Let $x = |K| + 1$ Since that guarantees $x > K$ always.

Sub $|K| + 1$ for x in x^2: $(|K| + 1)^2 \to K^2 + 2|K| + 1$

$$\left.\begin{array}{l} K^2 \text{ is always } 0 \text{ or } > 0 \\ 2|K| \text{ is always } 0 \text{ or } > 0 \\ 1 \text{ is always } > 0 \end{array}\right\} \text{all always greater than } K.$$

□ We have found an x $(|K| + 1)$ where $x^2 > K$, which contradicts our claim. Therefore, the original claim must be true.

4.)

If $\underbrace{3 \mid (n^2 + 2)}_{p} \to \underbrace{3 \nmid n}_{q}$

$p \to \sim q =$ If $3 \mid (n^2 + 2)$ then $3 \mid n$

★ $a \mid b$ iff $\exists c \in \mathbb{Z},\ ac = b$.

Using our definition, $3 \mid n = n = 3c$

Similarly, $3 \mid (n^2 + 2) = n^2 + 2 = 3d$

Since $n = 3c$, we can substitute $3c$ for n in $n^2 + 2$

$(3c)^2 + 2 = 3d \to 9c^2 + 2 = 3d \to 2 = 3d - 9c^2$

$2 = 3\underbrace{(d - 3c^2)}_{m} = \underline{\underline{2 = 3m}} =$ by definition is $3 \mid 2$

Which is false because $\frac{2}{3} \notin \mathbb{Z}$

□ Therefore the original claim must be true.